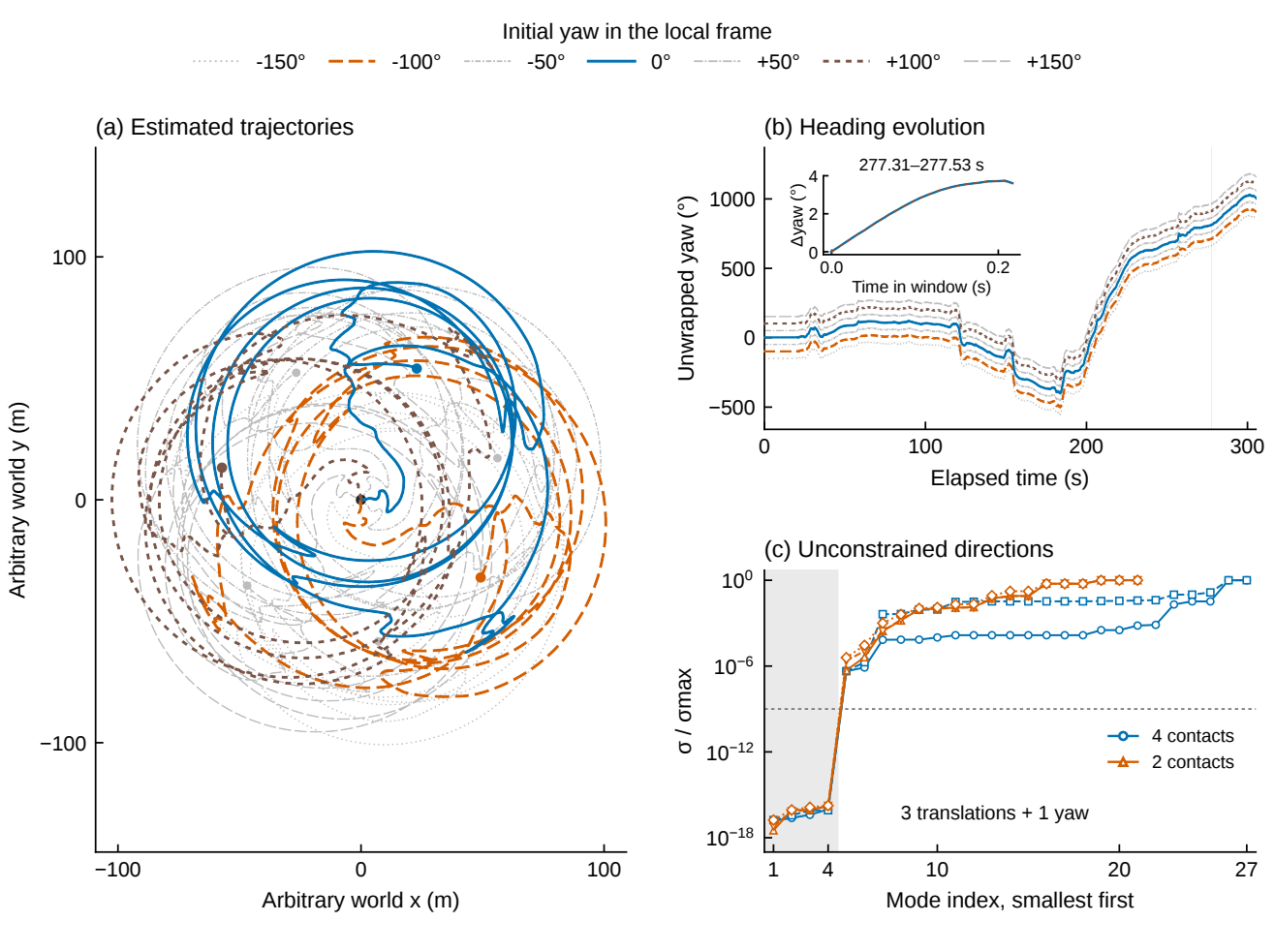

Initial yaw in the local frame
-150°
-100°
-50°
0°
+50°
+100°
+150°
(a) Estimated trajectories
Arbitrary world y (m)
Arbitrary world x (m)
100
0
−100
−100
0
100
(b) Heading evolution
Unwrapped yaw (°)
Elapsed time (s)
1000
500
0
−500
0
100
200
300
277.31–277.53 s
Δyaw (°)
Time in window (s)
4
2
0
0.0
0.2
(c) Unconstrained directions
σ / σmax
Mode index, smallest first
10^{0}
10^{-6}
10^{-12}
10^{-18}
1
4
10
20
27
4 contacts
2 contacts
3 translations + 1 yaw